Torus Solutions Beyond the GRW Assumption

A supplement to "Exact Λ-Vacuum Solutions with Curled Extra Dimensions"

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About this supplement

This supplement was generated by Claude, an AI assistant made by Anthropic. Bud Simrin's part was the series of questions that led to it. He did not check the derivations himself, but he ran the verification notebook, and its outputs match this document. The calculations were done by Claude in SymPy (Python) and Mathematica. The accompanying Mathematica notebook, Torus Solutions Beyond GRW.nb, repeats the key calculations so that readers can check them.

1. The question

In the main notebooks the curled torus cases were solved under the assumption $B = 1/(D_4 D_5 \cdots)$, the radial coefficient being the reciprocal of the product of the circle coefficients. With two circles (6D_1) this forces one circle to keep a constant size. With three circles (7D_1) all three can vary, and their exponents obey the Kasner conditions. Two questions follow:

- What would four circles (8D_1) add?
- What happens if B is not assumed in advance, but left free for the field equations to decide?

In all torus cases the field equations force $k = 0$. By the curvature test (Section 6.3 of the main report) the spatial metric h must then be Ricci-flat: $R_{ij}(h) = 0$.

2. Counting free parameters when $B = 1/(D_4 D_5 \cdots)$

With this assumption every torus case has the same solution,

$$D_i = (1 + \chi/r)^{e_i}, \quad \text{with} \quad \Sigma\, e_i = 1 \quad \text{and} \quad \Sigma\, e_i^2 = 1.$$

With n circles there are n exponents and 2 conditions. Geometrically the conditions describe a plane cutting a sphere in n-dimensional exponent space, which leaves an $(n - 2)$-dimensional sphere of solutions:

Circles	Case	Solutions for the exponents	Free parameters
1	5D	e = 1	0
2	6D_1	two points: (1, 0) and (0, 1)	0 (one circle constant)
3	7D_1	a circle (the Kasner circle)	1
4	8D_1	a 2-sphere	2

So 8D_1 adds one more free parameter and nothing qualitatively new. Smoothness at a pinch again picks one exponent equal to 1 and the others equal to 0 (Section 6.4 of the main report).

3. Leaving B free

Write $f = 1 - 2m/r$ (so $\chi = -2m$), and consider the spatial metric

$$h = f^{-s}\,dr^2 + f^{1-s}\,r^2\,(d\theta^2 + \sin^2\theta\,d\varphi^2) + \Sigma\, f^{e_i}\,p_i^2\,d\psi_i^2, \quad s = \Sigma\, e_i.$$

For two circles the Mathematica notebook finds that every component of $R_{ij}(h)$ vanishes except one, the radial component R_{11} (index 1 is r, as in the main notebooks):

$$R_{11} = -2m^2\,(e_4^2 + e_4e_5 + e_5^2 - 1) \,/\, [r^2\,(r - 2m)^2].$$

So h is a solution exactly when $e_4^2 + e_4e_5 + e_5^2 = 1$. Since $s = e_4 + e_5$, this can be written $s^2 + e_4^2 + e_5^2 = 2$, and in that form it holds for any number of circles: a numerical check with four circles gives $R_{ij} = 0$ to within rounding error (10^{-16}) when the condition holds, and not otherwise.

This is one condition instead of two, so freeing B gives one more free parameter. The assumption $B = 1/(D_4D_5\cdots)$ corresponds exactly to $s = 1$: then the angular coefficient is $f^0 = 1$ ($C = 1$, so r is the areal radius), $B = f^{-1}$, and the condition reduces to the Kasner conditions of Section 2.

For two circles the solutions now form an ellipse in the (e_4, e_5) plane:

- **$s = 1$:** the ellipse meets the line $e_4 + e_5 = 1$ at (1, 0) and (0, 1), the 6D_1 solutions.
- **$s \neq 1$:** every other point of the ellipse is a genuinely new solution. It is not an FG-GRW change of coordinates: written with the areal radius as coordinate, its D's are not simple powers and $B \neq 1/(D_4D_5)$.
- **One circle (5D):** the condition gives $e = \pm 1$, but $e = -1$ is the $e = 1$ solution with $\chi \to -\chi$, so nothing new appears.

4. The new solutions are singular

To test regularity, the notebook computes the Kretschmann invariant $K = R_{abcd}R^{abcd}$, a measure of curvature that does not depend on the choice of coordinates. For the 6D_1 solutions ($s = 1$) it is

$$K = 48\,m^2 \,/\, r^6,$$

which stays finite as $r \to 2m$ (the same value as for the Schwarzschild metric). For general exponents, K behaves like $(r - 2m)^{2s-4}$ times a polynomial in e_4 and e_5. On the ellipse $|s| \le 2/\sqrt{3} \approx 1.155$, so the power is negative, and the polynomial vanishes at $r = 2m$ only at (1, 0) and (0, 1) (found with Mathematica's Solve). Everywhere else on the ellipse, K becomes infinite at $r = 2m$. The point $r = 2m$ lies at a finite proper distance, so this is a genuine curvature singularity, not an artifact of the coordinates.

Point on the ellipse	K at r − 2m = 0.1	0.01	0.001	0.0001
s = 1, e = (1, 0) (6D_1)	6.8	11	12	12
s = 0.6, e ≈ (1.15, −0.55)	386	2.5×10^5	1.6×10^8	1.0×10^{11}
s = 1.1, e ≈ (0.85, 0.25)	5.8	112	6.8×10^3	4.3×10^5
s = −0.5, e ≈ (0.65, −1.15)	8.6×10^5	9.2×10^{10}	9.3×10^{15}	9.3×10^{20}

(m = 0.5, so r = 2m = 1 is where f = 0. All four points satisfy $e_4^2 + e_4e_5 + e_5^2 = 1$.)

Even at s = 1 the curvature being finite is not the whole story. The pinch is a smooth cap only when $\chi = -p/2$ (Section 6.2 of the main report); otherwise it is a conical tip, a much milder defect than a curvature singularity.

5. Conclusion

- Leaving B free enlarges the torus solutions by one parameter, already in 6D.
- Every new solution has a curvature singularity at r = 2m. The only nonsingular solutions are the s = 1 ones, and they are exactly the GRW solutions of the main notebooks.
- So the assumption $B = 1/(D_4D_5\cdots)$ is not merely a convenience. It selects precisely the solutions that can contain a smooth Kaluza–Klein bubble.
- Four or more circles add free exponents but no new kind of behavior.

Related literature

The five-dimensional version of this family, with time playing the role of one circle, is known in Kaluza–Klein theory; there the condition appears as $a^2 + ab + b^2 = 1$, which is the two-exponent condition of Section 3. The four-dimensional analog with a scalar field is the Fisher (Janis–Newman–Winicour) solution, which also has a naked singularity. The n-circle condition $s^2 + \Sigma\, e_i^2 = 2$ was checked directly here rather than taken from these sources.

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